

Sample Questions

2 Mark Questions

1. What is the purpose of a template file in Drupal?
2. Name any two fields typically filled while creating an article in Drupal.
3. Define a static page and give one example.
4. Mention two publishing options available while saving a static page.
5. What is the function of a content block in Drupal?
6. Define a "region" in Drupal with one example.
7. What does the "Demonstrate block regions" option show?
8. State any two methods to insert links into blocks.
9. What is the use of the "Manage display" setting in custom blocks?
10. What is the role of the Views module in Drupal?

6 Mark Questions

1. Explain the steps involved in creating an article in Drupal using templates.
2. Describe the process of creating and placing a custom content block in Drupal.
3. Discuss how static pages are created and managed in Drupal.
4. Explain how to reorder fields within a custom block type.
5. Describe how to create and configure a view in Drupal to display content.

10 Mark Questions

1. Describe the complete process of building a static page in Drupal, including publishing and URL alias settings.
2. Explain how content blocks, regions, and visibility settings work together to control content display in a Drupal website.
3. Discuss the various types of content that can be managed using Views in Drupal. Include the steps of creating a view and its display options.
4. Elaborate on the installation and configuration of a new module in Drupal. Include both file upload and URL installation methods.

LAB MANUAL

Lab 1

Install and Configure Drupal on Your Computer

Objective

To install and configure Drupal using XAMPP in a local environment for web development practice.

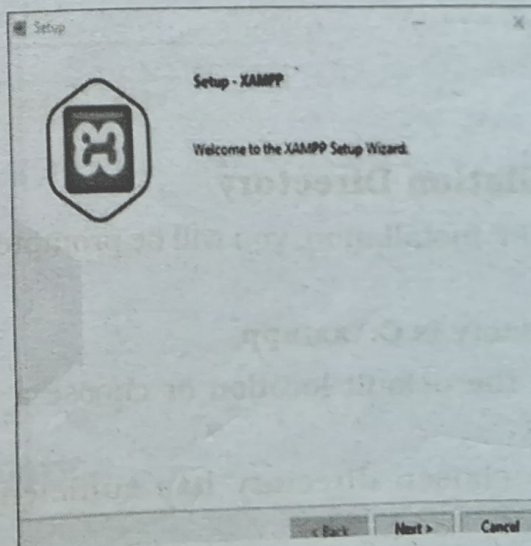
Procedure

Step 1: Downloading Drupal

- Visit the official Drupal website: <https://www.drupal.org/download>
- Download **Drupal 10.4.6**
- Save the downloaded archive file to your system for later use in the installation process

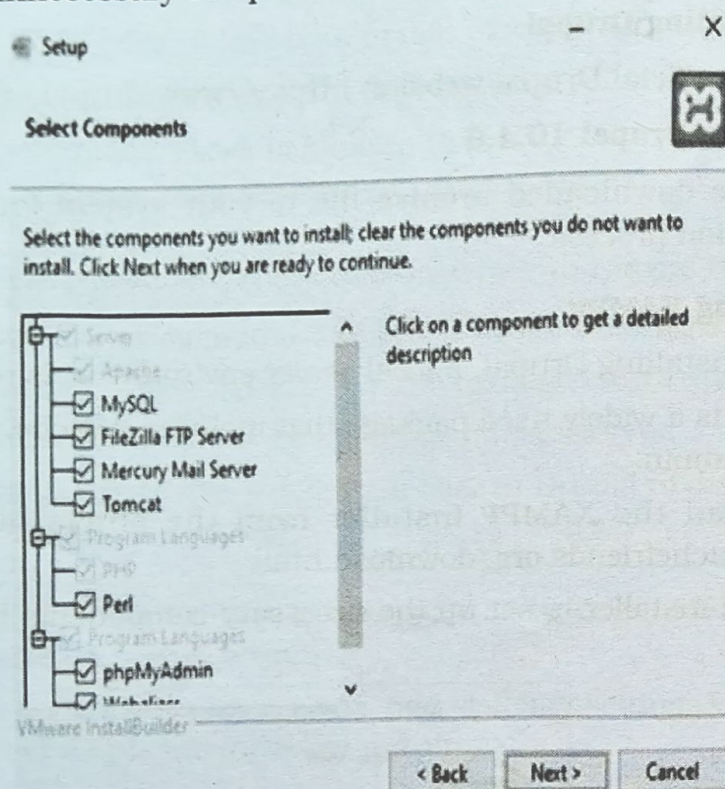
Step 2: Installing XAMPP

- Before installing Drupal, a local server environment is required.
- **XAMPP** is a widely used package that includes Apache, MySQL, PHP, and phpMyAdmin.
- Download the XAMPP installer from the official website: <https://www.apachefriends.org/download.html>
- Run the installer to set up the necessary components for hosting Drupal locally.



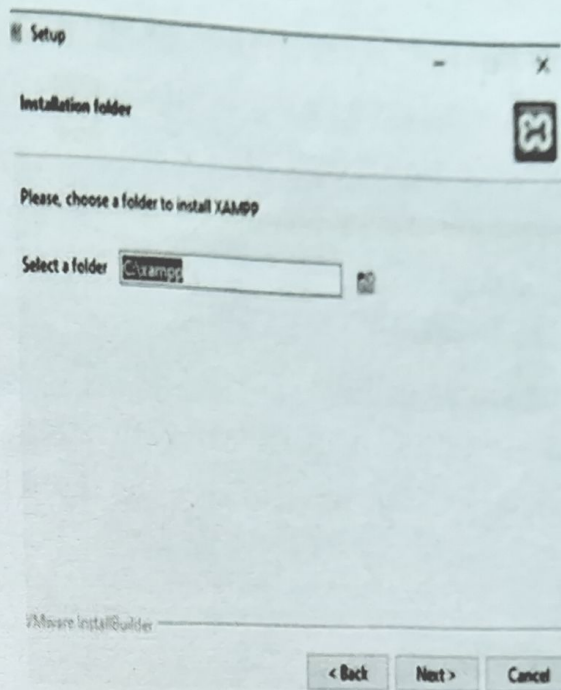
2.1 Selecting Components

- Begin the XAMPP installation by launching the setup file.
- In the component selection screen, choose only the necessary components.
- Recommended components for Drupal installation:
 - **Apache** (Web server)
 - **MySQL** (Database server)
 - **PHP** (Scripting language)
 - **phpMyAdmin** (Database management tool)
- Deselect unnecessary components to keep the setup lightweight.



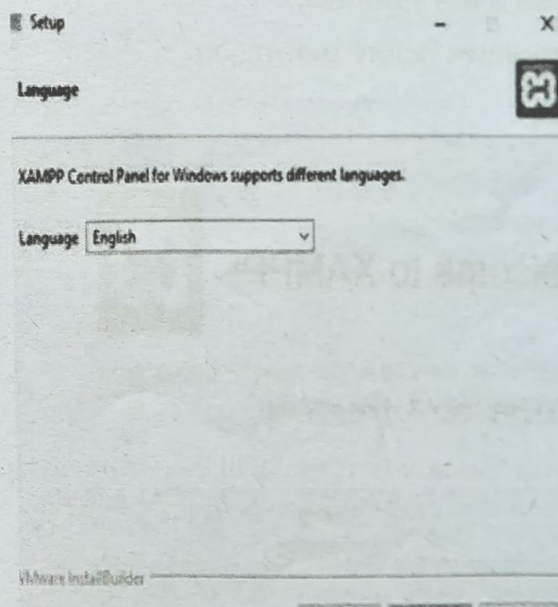
2.2 Choosing Installation Directory

- During the XAMPP installation, you will be prompted to select the installation folder.
- The default directory is **C:\xampp**.
- You can accept the default location or choose a different folder based on your preference.
- Make sure the chosen directory has sufficient disk space and write permissions.



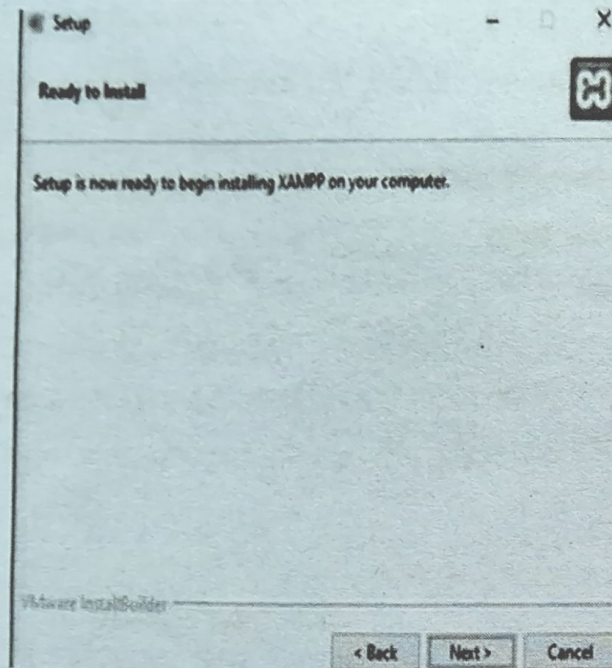
2.3 Selecting Language

- Choose **English** or your preferred language for the XAMPP installation interface.



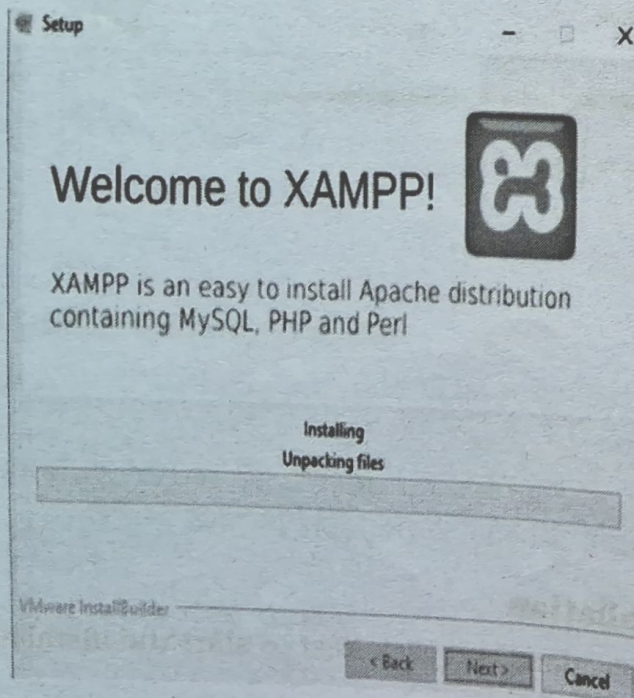
2.4 Beginning Installation

Review all your selections and click Next to start the installation process.



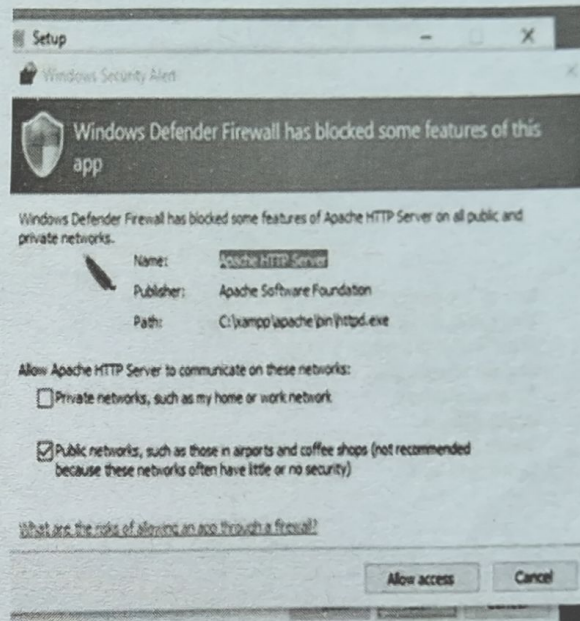
2.5 XAMPP Installation Started

- The installation begins automatically after confirmation.
- Necessary files and services like Apache and MySQL are set up.
- The process takes a few minutes.
- Wait until it completes before moving on.

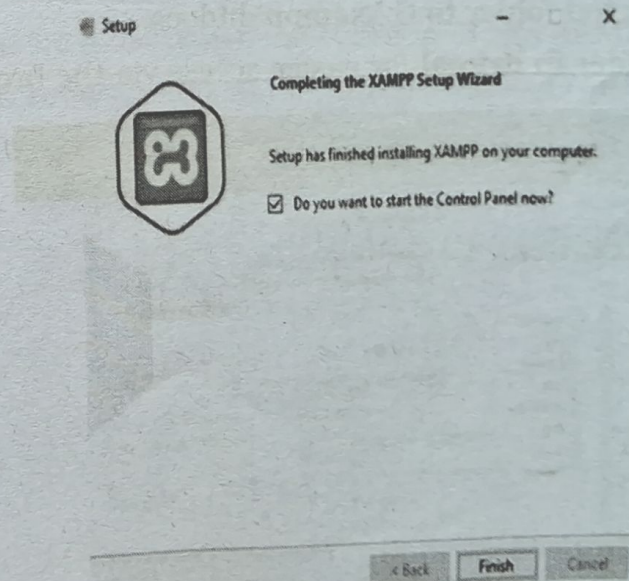


2.6 Completing the Installation

- If prompted by the firewall, click **Allow access** to permit Apache and MySQL services.



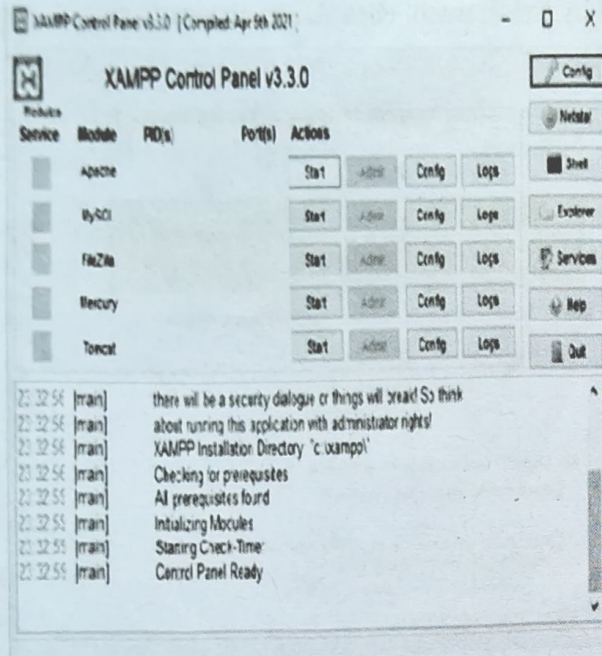
- After installation finishes, click **Finish** to open the XAMPP Control Panel.



2.7 Starting Server Service

- In the XAMPP Control Panel, click **Start** next to **Apache** to launch the web server.

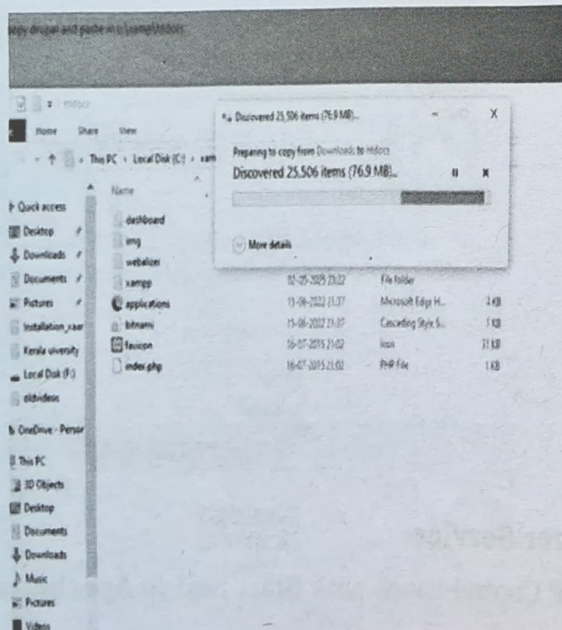
- Click **Start** next to **MySQL** to activate the database server.



Step 3: Setting Up Drupal in XAMPP

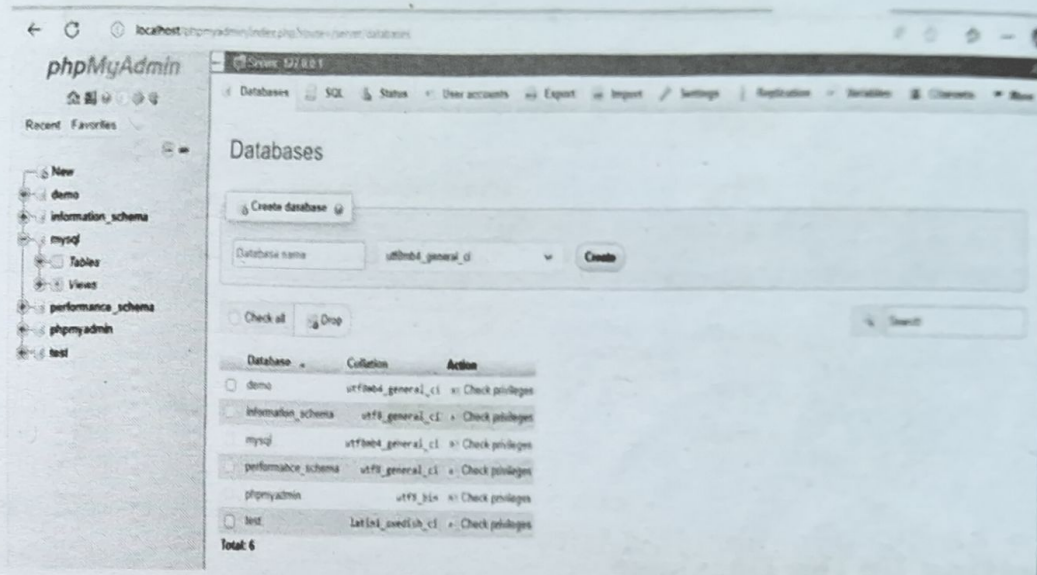
3.1 Copying Drupal Files

- Extract the downloaded Drupal archive.
- Copy the extracted folder to C:\xampp\htdocs.
- Rename the folder to **drupal** for easier access via the browser.



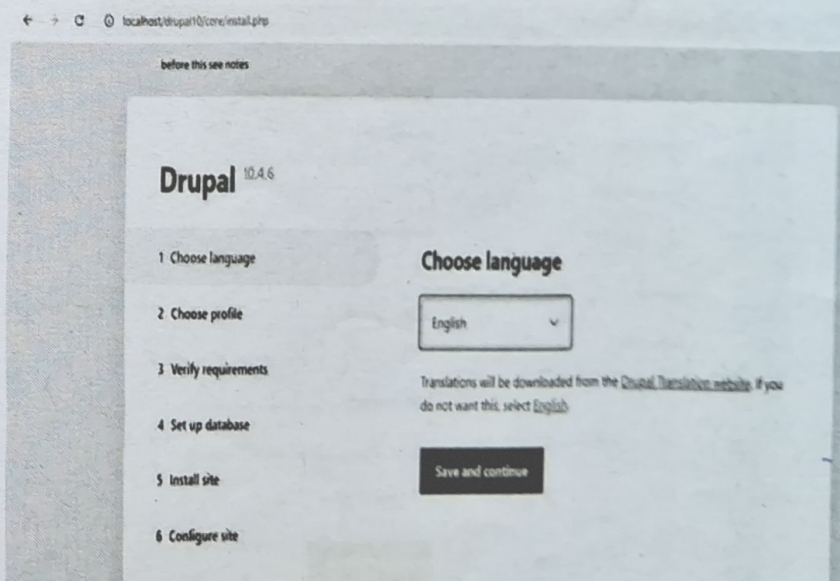
3.2 Creating the Database

- Open the XAMPP Control Panel.
- Click **Admin** in the MySQL section to open phpMyAdmin.
- Create a new database named **demo** to be used by Drupal.



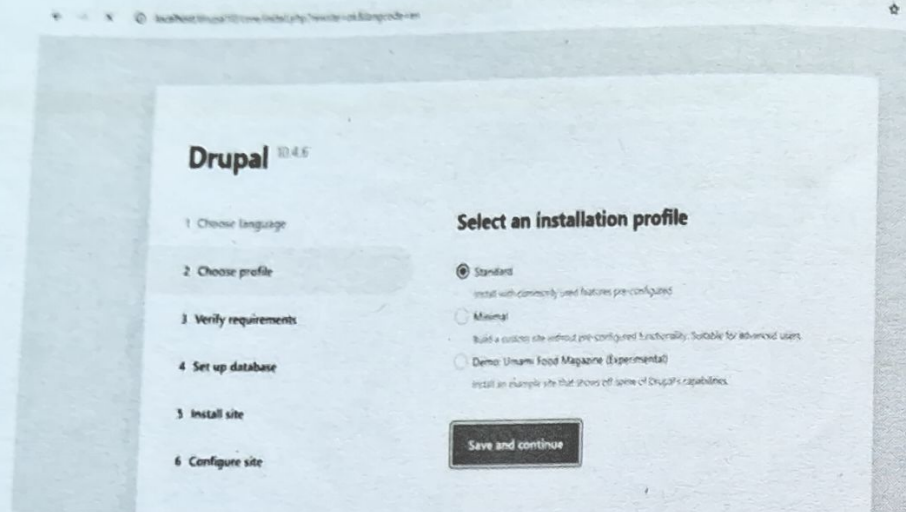
3.3 Running the Drupal Installer

- Ensure Apache and MySQL are running in the XAMPP Control Panel.
- Open a web browser and go to: <http://localhost/drupal>.
- The Drupal installation page will be displayed.



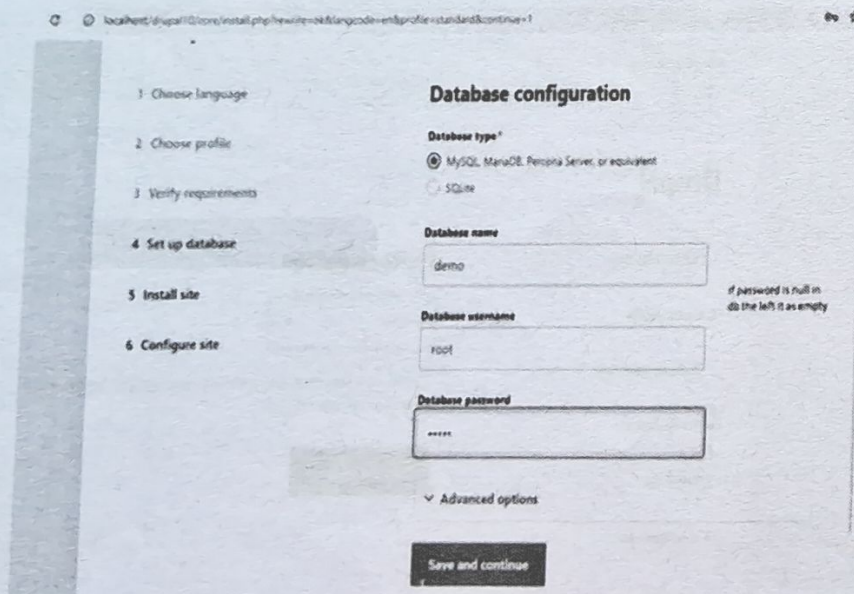
3.4 Choosing Installation Profile

- Select the **Standard** installation profile.
- This enables default modules and configurations suited for general websites.



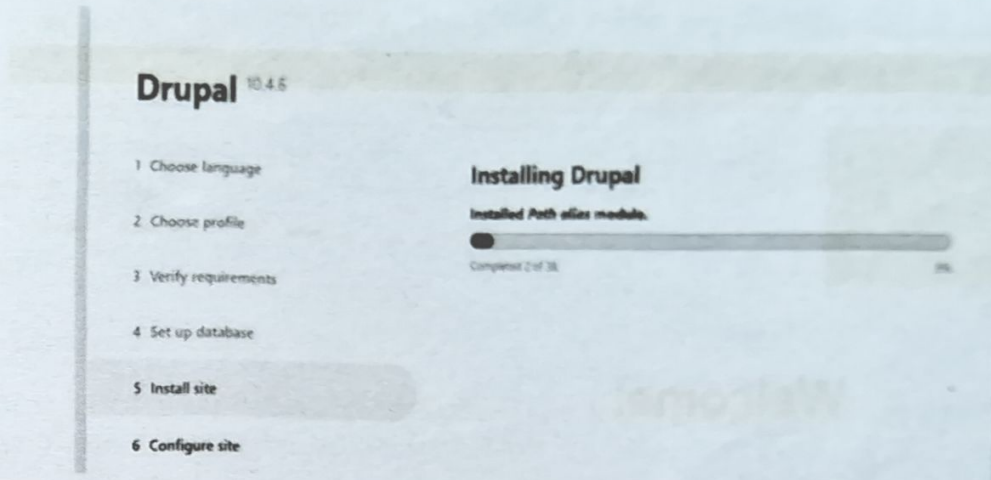
3.5 Setting Up the Database

- Enter the following database connection details:
 - **Database name:** demo
 - **Username:** root
 - **Password:** (leave blank — default in XAMPP)
- Click save and continue to begin the installation process.



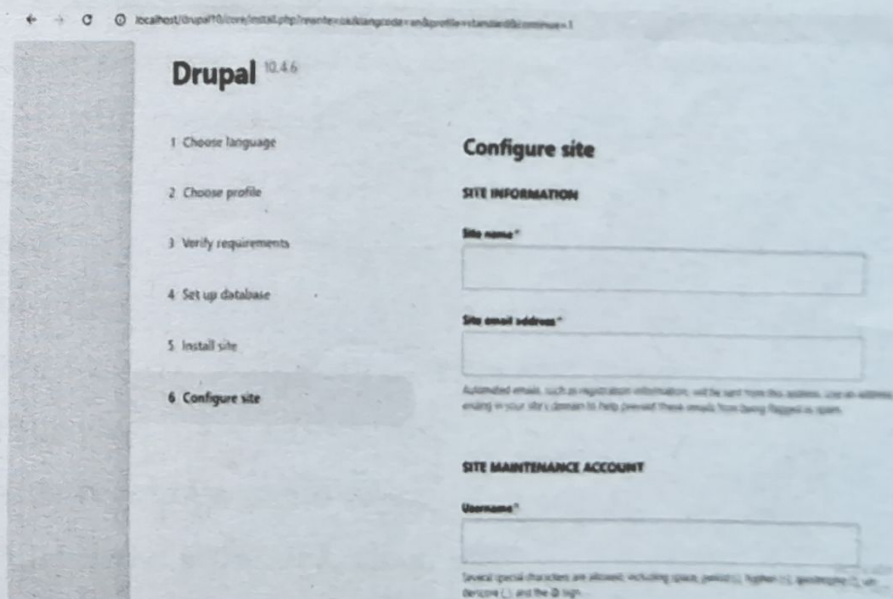
3.6 Drupal Installation Begins

- Drupal sets up the database and installs default modules.
- The process takes a few minutes with a progress bar displayed.



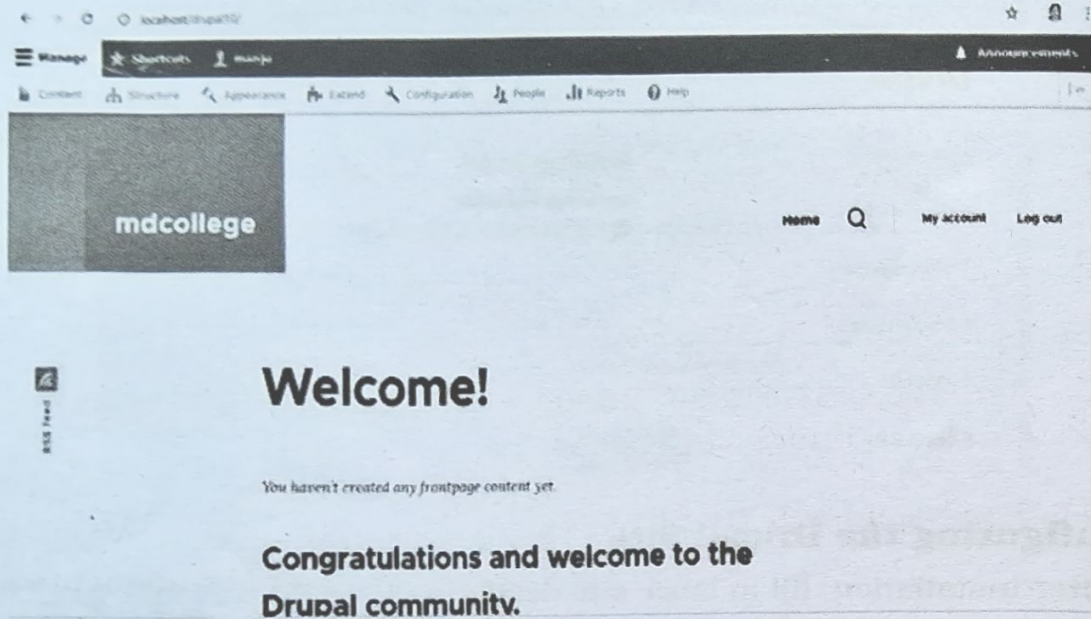
3.7 Configuring the Drupal Site

- After installation, fill in basic site details:
 - Site name
 - Administrator email
 - Admin username and password
 - Default language and time zone



3.8 Completion of Installation

- Once setup is complete, Drupal will display the site home page.
- The site is now ready for content creation and further customization.



Result

Drupal was successfully installed and launched locally, ready for content creation and customization.

Lab 2: Design a Website of Your College Using Drupal and Modify the Basic Site Settings

Objective

To design and build a basic college website using Drupal by creating pages, articles, and menus, and organizing them within the site structure using block placement.

Procedure

Step 1: Launch Drupal

Open your web browser and enter the following address to access the Drupal site: <http://localhost/drupal10>

Step 2: Identify Basic Website Elements

Explore the Drupal interface and identify key structural components such as:

- **Pages**
- **Menus**
- **Articles**

Step 3: Create a Basic Page

3.1 Navigate to: **Content** → **Add content**

3.2 Under content types, select:

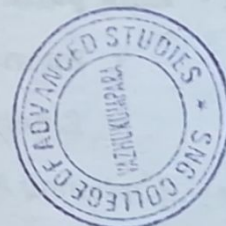
- **Basic page**
- **Article**

Select **Basic page**.

3.3 Fill in the required fields such as **Title** and **Body**.

3.4 To add the page to the site's menu:

- Enable **Provide a menu link**
- Specify **Menu title**, **URL alias**, etc.



3.5 Click **Save** to publish the page.

Step 4: Create an Article

4.1 Navigate to: **Content** → **Add content**

4.2 Under content types, select:

- **Basic page**
- **Article**

Select **Article**.

4.3 Enter details such as **Title**, **Body**, and upload an **Image**.

4.4 To link the article to a menu:

- Enable **Provide a menu link**
- Fill in the menu-related settings.

4.5 Click **Save** to publish the article.

Step 5: Create and Place a Menu

5.1 Navigate to: **Admin toolbar** → **Structure** → **Menus**

5.2 Click on “**Add Menu**”

Fill in the menu details:

- **Menu title** (e.g., *Quick Links*, *Footer Menu*, *My Custom Menu*)
- **Administrative summary** (optional)

5.3 Click **Save**

5.4 After saving, click “**Add link**”

5.5 Enter the following:

- **Menu link title** (e.g., *About Us*)
- **Link path:**
 - Internal: */about-us* or *node/5*
 - External: *https://example.com*
- (Optional) Add description or set a **parent link** for dropdowns

5.6 Click **Save**

5.7 To place the menu on the site:

- Go to: **Structure** → **Block layout**
- Identify a region (e.g., *Sidebar, Footer, Header*)
- Click "**Place block**"
- Search for your menu (e.g., *My Custom Menu*)
- Click "**Place block**" next to it

In the block configuration window:

- (Optional) Edit or hide the block title
- Set **visibility conditions** (e.g., display only on specific pages)
- Click **Save block**

Result

A college website was successfully created in Drupal, featuring custom pages, articles, and menus properly integrated and displayed in the designated theme regions.

Lab 3

Add Different Menus to Your Website in Drupal. The menu should contain Home, News, Gallery, About us and Contact us.

Objective

To create a custom News menu in Drupal and configure its links and placement within a specific region of the website layout.

Procedure

Step 1: Navigate to Menu Management

- Go to: **Admin toolbar '!' Structure '!' Menus**

Step 2: Create the “News” Menu

- Click on **“Add menu”**
- Fill in the following details:
 - **Menu title:** *News*
 - **Administrative summary:** *(optional – for internal use)*
 - Click **Save**

Step 3: Add a Link to the News Menu

- After saving the menu, click on **“Add link”**
- Enter the following link details:
 - **Menu link title:** *News*
 - **Link:** */news (internal path to the News page or article listing)*
 - **Parent link:** Select <Main Navigation> or another menu based on your preference
 - Click **Save**

Step 4: (Optional) Place the Menu on the Website

If you want to display the **News** menu in a specific region of the site:

4.1 Go to: **Structure → Block layout**

4.2 Identify the region where the menu should appear

(e.g., *Sidebar, Footer, Header*)

4.3 Click "**Place block**" next to the desired region

4.4 Search for your newly created menu (e.g., *News*)

4.5 Click "**Place block**" next to it

Step 5: Configure Block Display

In the block configuration window:

- (Optional) Change or hide the **block title**
- Set **visibility conditions** (e.g., display only on specific pages)

Click **Save block** to apply changes.

Note: Repeat the same process to create and manage other menus

Result

The News menu was successfully created, linked, and placed in a selected region of the Drupal site layout. Additional menus can now be created and placed using the same procedure.

Lab 4

Create User Roles for Your Site and Assign Permissions in Drupal

Objective

To learn how to add a new user in Drupal using administrative privileges and assign roles and credentials.

Procedure

Step 1: Log in as Administrator

- Open your Drupal site in a web browser.
- Log in using an account with **administrative privileges**.

Step 2: Navigate to the People Section

- From the **Admin toolbar**, go to:

Manage → People

Step 3: Add a New User

- Click the “+ **Add user**” button located at the top of the **People** page.
- Fill in the required user details:
 - **Email address**
 - **Username**
 - **Password**
 - **Assigned Role** (e.g., *authenticated user*, *editor*, *administrator*)
- Click **Save** to create the user account.

Note:

Once the account is saved:

- The new user can **log in** using the provided credentials.
- Alternatively, they can **reset their password** using the email notification sent to their registered email address.

Result

A new user was successfully created and can now access the Drupal site based on the role and permissions assigned.

Lab 5

Install and Activate a New Theme from the Drupal Theme Repository

Objective

To learn how to install a new theme in Drupal from the official theme repository and set it as the default theme for your site.

Procedure

Step 1: Log in as Administrator

- Open your Drupal site in a web browser.
- Log in using an account with administrative privileges.

Step 2: Navigate to the Appearance Section

- From the Admin toolbar, go to:

Manage → Appearance

Or visit directly: /admin/appearance

Step 3: Install a New Theme

- Click the **"Install new theme"** button at the top of the Appearance page.

Step 4: Provide the Theme Source

- You will be prompted to either:
 - Upload a **.tar.gz** theme archive file from your local system, or
 - Paste the **URL** of the theme from Drupal.org Theme Repository

Step 5: Install the Theme

- Click the **Install** button to begin the installation process.

Step 6: Enable and Set as Default

- After installation, locate the new theme in the list of available themes.
- Click **"Install and set as default"** to apply the theme to your site.

Result

A new Drupal theme was successfully installed and activated as the default theme for your site.

Lab 6

Add Different Blocks to Your Website in Drupal

Objective

To learn how to create a custom block in Drupal and place it in a specific region of your site layout.

Procedure

Step 1: Log in as Administrator

- Open your Drupal site in a web browser.
- Log in using an account with administrative privileges.

Step 2: Navigate to the Block Layout Section

- From the Admin toolbar, go to:

Manage → Structure → Block layout

Step 3: Create a Custom Block

- Click on the “+ Add custom block” button.

Step 4: Fill in the Block Details

- **Block description:**

Welcome Message

- **Body (use the rich text editor):**

Type the welcome text you want to display.

- Click **Save** to create the custom block.

Step 5: Place the Custom Block

- Return to the **Block layout** page.
- Under a desired region (e.g., **Sidebar First**), click the “**Place block**” button.

- Search for the custom block labeled **Welcome Message**.
- Click "**Place block**" next to it.

Step 6: Configure Block Settings

- Optionally, adjust the **Visibility settings** (e.g., display only on the homepage).
- Click **Save block** to apply changes.

Result

A custom block titled Welcome Message was successfully created and placed in the chosen region of your Drupal site.

Lab 7

Create a New Content Type and Add Some Fields

Objective

To learn how to create a new content type in Drupal and add custom fields to structure specific types of content such as events, articles, or announcements.

Procedure

Step 1: Log in as Administrator

- Open your Drupal site in a web browser.
- Log in using an account with administrative privileges.

Step 2: Navigate to the Content Types Section

- From the Admin toolbar, go to:

Manage → Structure → Content types

Step 3: Add a New Content Type

- Click the blue **+ Add content type** button at the top of the page.

Step 4: Fill in the Content Type Details

- **Name:** Enter a name for your new content type (e.g., Event, News Article).
- **Description (optional):** Add a short description (e.g., Used to publish upcoming events on the college website).

Step 5: Configure Optional Settings

- Expand and review the following sections:
 - ✓ **Submission form settings:** Change field labels if needed (e.g., rename "Title" to "Event Name").
 - ✓ **Publishing options:** Choose whether new content is published by default, promoted to the front page, or marked as sticky.
 - ✓ **Display settings:** Decide whether to show the author and post date.

- ✓ **Menu settings:** Enable the option to include this content type in site menus if required.

Step 6: Save the Content Type

- Click "**Save and manage fields**" to continue to the field configuration section.

Step 7: Add a Custom Field

- Click the "+ **Add field**" button.
- Choose from the two options:
 - Re-use an existing field (shared field), or
 - Add a new field

Step 8: Choose the Field Type and Configure

- Under **Add a new field**, select a field type from the dropdown list:
 - **Text (plain):** For short input like titles or locations
 - **Text (formatted, long):** For detailed content or descriptions
 - **Image:** To allow image uploads
 - **Date:** For events or deadlines
 - **Link:** To add external or internal URLs
 - **Boolean:** For checkboxes (e.g., "Is it a featured event?")
 - **List (text):** For dropdown options or multiple choices
- Provide a **Label** for the field (e.g., Event Date)
- The **Field name** will auto-generate (e.g., field_event_date)
- Click "**Save and continue**" and follow prompts to configure additional field settings.

Result

A new content type was successfully created, and custom fields were added to support structured content creation in Drupal.

Lab 8

Add a New Article to Your Site

Objective

To learn how to create and publish a new article in Drupal using the built-in Article content type.

Procedure

Step 1: Log in as Administrator

- In your Drupal site in a web browser.
- Log in using an account with administrative privileges.

Step 2: Navigate to the Add Content Section

- From the Admin toolbar, go to:

Manage → Content → + Add content → Article

Step 3: Fill in Article Details

- Provide the following information:
 - ✓ **Title:** New Invention
 - ✓ **Body:** Type the content of your article (e.g., description, background, or summary of the invention).
 - ✓ **Image (optional):** Upload a related image or diagram that supports the article content.

Step 4: Set Publishing Options

- Scroll to the **Publishing options** section and configure as needed:
 - ✓ **Published:** Checked
 - ✓ **Promoted to front page:** Checked (if you want the article on the home page)

- × **Sticky at top of lists:** Optional, leave unchecked unless needed

Step 5: Save the Article

- Click the **"Save"** button at the bottom of the page to publish the article.

Result

A new article titled New Invention was successfully created and published on your Drupal site.

Lab 9

Install and Configure the 'Pathauto' Module

Objective

To install and enable the Pathauto module in Drupal, which automatically generates SEO-friendly URL aliases for your site's content.

Procedure

Step 1: Log in as Administrator

- Open your Drupal site in a web browser.
- Log in using an account with administrative privileges.

Step 2: Navigate to the Module Installation Page

- From the Admin toolbar, go to:

Manage → Extend → + Install new module

Step 3: Download the Pathauto Module

- Visit the official Pathauto module page:

<https://www.drupal.org/project/pathauto>

- Copy the download link for the .tar.gz or .zip file.
- Paste the link into the "**Install from a URL**" field on your Drupal site.
- Click the **Install** button to begin installation.

Step 4: Enable the Pathauto Module

- After installation, return to the **Extend** section:

Manage → Extend

- In the search bar, type **Pathauto**.
- Check the box next to the **Pathauto** module.
- Click the **Install** button at the bottom of the page to enable the module.

Result

The Pathauto module was successfully installed and enabled, allowing Drupal to automatically generate clean, SEO-friendly URLs for your site's content.

Lab 10

Create the Mobile View of Your Website

Objective

To learn how to preview and test how your Drupal website appears and behaves on various mobile devices using browser developer tools.

Procedure

Step 1: Open Your Drupal Site in Google Chrome

- Launch the **Google Chrome** browser.
- Open your Drupal site by entering the site URL in the address bar.

Step 2: Open Chrome Developer Tools

- Right-click anywhere on your Drupal site and select **Inspect**,

or

- Press **Ctrl + Shift + I** (Windows)
- Press **Cmd + Option + I** (Mac)

Step 3: Enable Device Toolbar for Mobile View

- In the Developer Tools panel, click the  **Toggle Device Toolbar** icon in the top-left corner,

or

- Press **Ctrl + Shift + M** (Windows)
- Press **Cmd + Shift + M** (Mac)

Step 4: Select a Mobile Device

- At the top of the page preview, locate the device dropdown menu (default is usually **Responsive**).

- Click the dropdown and choose from a list of popular mobile devices, such as:
 - iPhone SE, iPhone XR, iPhone 12 Pro
 - Google Pixel 7, Samsung Galaxy S9+
 - iPad, iPad Pro

Step 5: Review the Site's Mobile Appearance

- Observe how your Drupal site adapts to the selected screen size and resolution.
- Check for responsive layout behavior, navigation usability, and readability on mobile screens.

Result

The mobile version of your Drupal site was successfully previewed using Chrome's developer tools, allowing for responsive design testing across various devices.

SYLLABUS

Module	Unit	Content	Hrs (30+30)	Marks
I		Introduction to Web Designing	6	10
	1	Basics of Web Designing -World Wide Web (WWW), W3C, Web Browser	1	
	2	Web Server, Web Hosting, Web Pages, DNS, URL	2	
	3	Overview of HTML (Concept only) and its role in Web Development	1	
	4	Open-Source S/W, Open-Source vs Closed Source Software, Open Source Licenses (Concept only)	2	
II		Introduction to CMS	6	10
	5	Introduction to Content Management Systems (CMS) - Features of CMS	2	
	6	Web Content Management System	2	
	7	Components of Content Management System	2	
III		Introduction to Drupal	8	15
	8	Drupal - Features, Advantages and Disadvantages, Installation and Configuration	2	
	9	Content types and Field Drupal Architecture	2	
	10	User Management, Managing Comments	2	
	11	Creating and Customizing Themes	2	
IV		Building Website	10	15
	12	Website Development - Working with Templates and Template files	2	
	13	Articles, Creating Web Forms	1	